

Institute of Actuaries of India

Subject CT2 – Finance and Financial Reporting

October 2015 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

<u>Solution 1:</u>	C	[2]
<u>Solution 2:</u>	C	[2]
<u>Solution 3:</u>	D	[2]
<u>Solution 4:</u>	C	[2]
<u>Solution 5:</u>	A	[2]
<u>Solution 6:</u>	B	[2]
<u>Solution 7:</u>	D	[2]
<u>Solution 8:</u>	C	[2]
<u>Solution 9:</u>	D	[2]
<u>Solution 10:</u>	B	[2]
<u>Solution 11:</u>		

i)

Statement of comprehensive income for the year ended on 31st March 2015

	Amounts in INR million			Marks
Revenue			20,000	
Cost of Sales				
Opening inventory	2,450			
Purchases	11,500			
(-)Closing inventory	(1,600)	12,350		0.5 for Closing inventory
Wages				
Balance as on 31st March 2015	3,500			
(+) Prepaid as on 31st March 2014	124			0.5 for prepaid balance
(-)Prepaid as at 31st March 2015	(100)	3,524		0.5 for prepaid balance
Power and fuel expenses		1,450		
Depreciation on Machinery and equipment		1,130		0.5 for depreciation
Depreciation on Factory building		240	(18,694)	0.5 for depreciation
Gross profit			1,306	
<u>Sales and distribution expenses</u>				
Depreciation on Motor vehicles		8		0.5

Advertisement expenses		250		
Distribution expenses		750	(1,008)	
-				
Administrative Expenses				
Salaries to administrative staff				
Balance as on 31st March 2015	400			
(+) Outstanding as on 31st March 2015	40	440		0.5 for outstanding balance
Printing and stationery		12		
Insurance Premium		15	(467)	
Discount received on payment to supplier			1	1.0
Operating Profit			(168)	
Finance Cost- outstanding interest on secured debentures			(320)	1.0
Profit before tax			(488)	
Dividend paid to Preference shareholders			NIL	0.5 only if written as nil
Total comprehensive income			(488)	

Notes to accounts:

Preference dividend has not been allowed for as the company has suffered losses

[10]

ii)

Statement of financial position as on 31st March 2015

	Amounts in INR million		Marks
<u>Non-current assets</u>			
Factory land		1,005	0.5
Factory building	2,400		0.5
(-) Depreciation for the year	(240)	2,160	0.5 for depreciation
Machinery and equipment	5,650		0.5
(-) Depreciation for the year	(1,130)	4,520	0.5 for depreciation
Motor Vehicles	40		0.5
(-) Depreciation for the year	(8)	32	0.5 for depreciation
Investments		1,150	0.5
<u>Current Assets</u>			
Inventory		1,600	0.5
Trade Receivables		3,000	0.5
Bills receivable	450		0.5

(+) Bill returned by supplier on 31st March 2015	40	490	0.5 for adding the bill returned
Prepaid wages		100	0.5
Cash and bank balances	554		0.5
(-) Cash paid to supplier in return of bills of exchange returned	(39)	515	0.5 for correctly subtracting cash paid after discount
Total Assets		14,572	
Equity and Liabilities			
Equity share capital		1,000	0.5
12% Preference share capital		500	0.5
Reserves and retained earnings	6,950		0.5
(+) Profits/(loss) for the current year	(488)	6,462	0.5 for adding balances
Total Equity		7,962	
Non-current liabilities			
8% Secured debentures		4000	0.5
Outstanding Interest		320	0.5
Current Liabilities			
Outstanding salary of administrative staff		40	0.5
Trade payable	2250		0.5
(+) Bills of exchange returned	40		0.5
(-) Cash paid in lieu of bill returned	(39)		0.5
(-) Discount received on cash payment in lieu of bill returned	(1)	2250	0.5 for subtracting discount received
Total equity and liabilities		14,572	

[13]

[23 Marks]

Solution 12:

- i) Net gain after tax per share if shares were sold today
 $= (319.75 - 250) \times (1 - 20\%) = \text{INR } 55.8 \text{ per share}$

Expected Net gain after tax per share if shares were sold at INR 390 on 1st February 2015
 $= (390 - 250) \times (1 - 10\%) \times 40\% = \text{INR } 50.4 \text{ per share}$

Expected Net gain after tax per share if shares were sold at INR X on 1st February 2015
 $= 55.8 - 50.4 = \text{INR } 5.4$

Since the two timing options have been mentioned as equal in value, X must be higher than the purchase price. Hence there will be a Capital Gain even if share prices go to X on 1st February 2015

Expected Net gain after tax per share if shares were sold at INR X on 1st February 2015
 $= (X - 250) \times (1 - 10\%) \times 60\% = 0.54 X - 135$

Hence X = INR 260 [3]

- ii) If the expected value of X is expected to be higher than INR 260, then Mr. Spooky should wait till 1st February to sell his shares

[1]

- iii) **This has two alternative solutions. The alternative solutions have been laid out as Option 1 and 2 below**

Option 1: If there is no other gain in the current financial year to set off the losses

Expected net gain after tax per share if shares were sold at INR 390 on 1st February 2015
 $= (390 - 250) \times (1 - 10\%) \times 40\% = \text{INR } 50.4 \text{ per share}$

Expected net gain after tax per share if shares were sold at INR 240 on 1st February 2015
 $= (240 - 250) \times 60\% = \text{INR } -6 \text{ per share}$

Hence Total Expected Gain = $50.4 - 6 = \text{INR } 44.4 \text{ per share}$

If the current price is assumed to be Y, net gain after tax per share if shares were sold today
 $= (X - 250) \times (1 - 20\%) = \text{INR } (0.8 X - 200)$

Hence X = INR 305.5 per share

Option 2: If there are other gains in the current financial year to set off the losses

Expected net gain after tax per share if shares were sold at INR 390 on 1st February 2015
 $= (390 - 250) \times (1 - 10\%) \times 40\% = \text{INR } 50.4 \text{ per share}$

Expected net loss after tax per share if shares were sold at INR 240 on 1st February 2015
 $= (240 - 250) \times (1 - 10\%) \times 60\% = \text{INR } 5.4 \text{ per share}$

Hence Total Expected Gain = $50.4 - 5.4 = \text{INR } 45 \text{ per share}$

If the current price is assumed to be Y, net gain after tax per share if shares were sold today
 $= (X - 250) \times (1 - 20\%) = \text{INR } (0.8 X - 200)$

Hence X = INR 306.25 per share

[2]

iv) Other Factors to be considered

- Future prospects of Global Ltd. If the company is expected to do well in a medium to long time horizon, then Mr. Spooky could hold the shares to benefit from the company's prosperity
- Risk averseness to the expected fall in share prices
- Broader economic factors (local and international) and expected changes in these factors in the next 16 day period as these could cause large shifts in the stock markets
- Need for immediate cash and ability to arrange for temporary funding at reasonable cost if required.
- Alternative investment avenues available and expected returns in these avenues. Whether any attractive investment avenue is open only for a short while and shall not be available till 1st February 2015
- Likelihood of any sizable dividend being declared in the near future

[2]

v) 3 alternative solutions have been listed below.

Alternative 1:

Buy a put option with strike price of close to INR 319.75

Benefit: Potential to gain from upside retained although the downside has been hedged

Drawback: Option premium shall be payable which may cause cash strain now

Challenges: Put options on shares of Global Ltd. may not be available

Even if they are available, it is difficult to hedge exposure to 13322 shares completely as options are generally available in fixed lot sizes of 10, 50, 100 or more

Alternative 2:

Sell a futures contract with strike price of close to INR 319.75

Benefit: Exchange traded and hence no counterparty risk and exposure can be easily offloaded

Drawback: Potential to gain from upside lost

Margins may have to be paid during the interim period causing cash strain

Challenges: Futures on shares of Global Ltd. may not be available

Even if they are available, it is difficult to hedge exposure to 13322 shares completely as futures are generally available in fixed lot sizes of 10, 50, 100 or more

Alternative 3:

Sell a forward contract with strike price of close to INR 319.75

Benefit: tailor made to hedge exposure on exactly 13322 shares of Global Ltd.

Drawback: Counterparty risk exists as it is not exchange traded

Potential to gain from upside lost

Challenges: Difficulty in finding a counterparty interested in entering into the required forward contract

[3]

[11 Marks]

Solution 13:

i) The Board should offer a share buyback to the existing equity shareholders

How it helps meet desired objectives:

- Cash pile offloaded
- Reduction in equity will increase gearing
- Two key stakeholders can invest in other opportunities available with cash received

[2]

ii) Share buyback could be achieved in the following ways:

- A fixed price offer
- A tender price (either a dutch or uniform price)
- Repurchase by direct negotiation with a major shareholder

[3]

iii) Taxation of Capital gains is an important consideration for minority shareholders to consider whether to offer their shares for buyback

[1]

[6 Marks]**Solution 14:**

i) And ii)

Discount Rate	9%				Amounts in Rs. Million		
Deposit Premium	15%						
		Years					
Without Reinsurance	0	1	2	3	4	5	6
Reserve Held	100.00	100.00	67.20	47.12	35.04	27.98	24.12
Claims Payment	0.00	40.00	25.00	15.63	9.77	6.10	3.51
Interest Earned	0.00	7.20	4.92	3.54	2.71	2.24	2.01
Reserves at end of year	0.00	67.20	47.12	35.04	27.98	24.12	22.63
Insurers Result (NPV)	13.49	$= (22.63) / (1.09)^6$					
With Reinsurance							
RI Premium	90.00						
Reserve with Insurer	10.00	This will grow to 16.77 at the end of 6 th year. i.e. $10 * (1.09)^6$					16.77
Deposit Premium	13.50	This will grow to 22.64 at the end of 6 th year i.e. $13.5 * (1.09)^6$					22.64
Total Reserve Available in Beginning		76.50	41.59	19.20	4.60	-5.19	-12.03
Claims Paid		40.00	25.00	15.63	9.77	6.10	3.51
Interest Earned		5.09	2.62	1.03	-0.03	-0.74	-1.24
Reserves at end of year		41.59	19.20	4.60	-5.19	-12.03	-16.78
Insurers Results (NPV)	10.00	$= (16.77)/(1.09)^6$					
Reinsurers Result (NPV)	3.49	$= (22.64 - 16.78) / (1.09)^6$					

Conclusion: Since insurer's NPV under the reinsurance option is lower than the NPV achieved by not reinsuring its portfolio, XYZ shouldn't reinsure its liability insurance portfolio

[5+7=12]

iii)

The negotiated premium will be total reserve amount minus the insurers PV = $100 - 13.49 = \text{Rs. } 86.51 \text{ Million}$.

Another approach would be to subtract the Reinsurer result of Rs. 3.49 Million from the RI premium of Rs. 90 Million to arrive at Rs. 86.51 Million.

Alternate approach will be to discount the claims paid over the years which will be Rs. 86.51 Million.

[2]

iv)

RI Premium Paid	-90.00
Total Claims Paid Recovered	100.00
Total Interest credited	-6.72
Result	3.28

According to CFO's analysis - There is a profit result of Rs. 3.28 Million.

This analysis is not suitable because this does not take into consideration the interest loss as a result of upfront payment of 15% Deposit Premium.

[2]

v)

Loss of interest on Deposit Premium = Rs. 22.64 Million – Rs. 13.5 Million = Rs. 9.14 Million

RI Premium Paid	-90.00
Total Claims Paid Recovered	100.00
Total Interest credited	-6.72
Loss of interest on Deposit Premium	-9.14
Result	-5.86
Discounted for 6 years at 9%	
Insurers Results (PV)	-3.49

[6]

[22 Marks]

Solution 15:

i)

Under the Modigliani & Miller assumption, weighted average cost of capital of both the companies should be the same since they are identical companies.

[1]

ii)

Company B

$$\begin{aligned}
 \text{Cost of Equity} &= R_f + \text{Beta} * (R_m - R_f) \\
 &= 6\% + 1.5 * (12\% - 6\%) \\
 &= 6\% + 9\% \\
 &= 15\%
 \end{aligned}$$

$$\begin{aligned}
 \text{Net Cost of Debt} &= \text{Gross Cost of Debt} * (1 - \text{Tax rate}) \\
 &= 7\% * (1 - 40\%) \\
 &= 4.2\%
 \end{aligned}$$

$$\begin{aligned}
 \text{Weighted average cost of capital} &= \text{Cost of Equity} * \text{Equity Proportion} + \text{Cost of Debt} * \text{Debt Proportion} \\
 &= 15\% * 2/3 + 4.2\% * 1/3 \\
 &= 11.4\%
 \end{aligned}$$

Company A

Total Debt = 400 Million

$$\begin{aligned}
 \text{Weighted Average Net Cost of Debt} &= (8\% * 40/400 + 9\% * 160/400 + 11.2\% * 200/400) * (1 - 40\%) \\
 &= 6\%
 \end{aligned}$$

$$\begin{aligned}
 \text{Weighted average cost of capital} &= \text{Cost of Equity} * \text{Equity Proportion} + \text{Cost of Debt} * \text{Debt Proportion}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thus} \quad 11.4\% &= \text{Cost of Equity} * 1/2 + 6\% * 1/2 \\
 \text{Cost of Equity} &= (11.4\% - 3\%) * 2 \\
 &= 16.8\%
 \end{aligned}$$

$$\begin{aligned}
 \text{Cost of Equity} &= R_f + \text{Beta} * (R_m - R_f) \\
 16.8\% &= 6\% + \text{Beta} * (12\% - 6\%) \\
 \text{Beta} &= (16.8\% - 6\%) / 6\% \\
 &= 1.8
 \end{aligned}$$

[5]

iii)

Company A desires to have Beta of 1.5

$$\begin{aligned}
 \text{Thus the required cost of equity} &= R_f + \text{Beta} * (R_m - R_f) \\
 &= 6\% + 1.5 * (12\% - 6\%) \\
 &= 15\%
 \end{aligned}$$

Net Cost of Debt = 6% (From Part I above)

Let the amount of additional equity required be X Million

$$\begin{aligned}
 \text{Weighted average cost of capital} &= \text{Cost of Equity} * \text{Equity Proportion} + \text{Cost of Debt} * \text{Debt Proportion}
 \end{aligned}$$

$$11.4\% = 15\% * (400 + X) / (800 + X) + 6\% * 400 / (800 + X)$$

$$X = 200 \text{ Million}$$

Hence 200 Million of additional equity should be raised making it a total of 600 Million of Equity.

[4]

[10 Marks]

Solution 16:**i) Liquidity preference-**

The liquidity preference theory is based on the generally accepted belief that investors prefer liquid assets to illiquid ones. Investors require a greater return to encourage them to commit funds for a longer period. Long-dated stocks are less liquid than short-dated stocks, so yields should be higher for long-dated stocks.

Appropriate examples-

1. Government bonds are usually more marketable and hence more liquid than corporate bonds and therefore corporate bonds have a higher yield to encourage investors to buy relatively illiquid assets.
2. 10 year bonds are relatively less liquid than One year bonds and hence there is a higher yield on 10 year bonds than on One year bonds.

[4]

ii) Financial Gearing-

The expression “gearing” or “financial gearing” is often used to refer to the impact on the profits for a company caused by fixed-interest borrowing. For a financially highly geared company a small change in the total profits might have a very large proportionate impact on the profits for shareholders. A company with lots of fixed-interest borrowing is “highly geared”.

Appropriate examples-

If the company/firm has positive outlook or is bullish on its performance, then to maximise its return on equity will choose to be highly geared. A highly geared company say 2 times will have return on equity higher than its return on capital, i.e. for example-

Equity	= 100 Million
Debt@ 6%	= 200 Million
Total Capital	= 300 Million
Return on capital	= 8%
Total returns	= 24 Million
Less Interest cost	= - 12 Million
Total returns on equity	= 12 Million
Return on Equity %	= 12%

And vice versa, a company which has doubts/ negative outlook/ concerns on its performance should lower the financial gearing to safeguard its return on equity.

[4]

[8 Marks]
